

Structuring College Access: The Market Segment Model and College Board Geomarkets

1 Data and Methods

1.1 Research Design and Case Selection

We utilize a quantitative case study design in which metropolitan areas are the cases. Metropolitan areas are the appropriate unit of analysis because Geomarkets partition metropolitan space into proprietary recruiting territories. Our empirical goal is to reconstruct how Geomarket filters within the College Board Student Search Service shape the racial/ethnic and first-generation composition of academically similar prospect pools made visible to colleges. We do so by analyzing actual student-list purchases and varying Geomarket inclusion and exclusion while holding academic search criteria constant. We treat Student Search Service as a market device whose built-in filters define a menu of possible actions. The analyses therefore focus on the distributional consequences of one filtering affordance built into the product

Our research questions require observing all Geomarkets within a metropolitan prospect pool in order to reconstruct how alternative Geomarket selections alter the composition of visible prospects. We restrict analyses to metropolitan areas for which we have sets of student-list purchases that (A) include every Geomarket in the metropolitan area, (B) define prospect pools primarily by high school graduating class and test-score range, and (C) repeat that structure across low, middle, and high test score bands. Although the primary analyses focus on prospects in the middle test score band, we report parallel results for the low and high bands in the Online Appendix, allowing readers to assess whether the observed patterns in racial and class composition hold across test

score ranges. We exclude purchases that layer on additional search criteria, such as intended undergraduate major or preferred college size, that produce dramatic and non-random reductions in the number of prospects. The final case set includes the Bay Area, Chicago, Dallas, Detroit, Long Island, Los Angeles, Northern New Jersey, Orange County, Philadelphia, and San Diego. We lack the requisite student list purchases to answer the three research questions for Atlanta, Boston, Cleveland, DC/Maryland/Virginia, Houston, Miami, and New York City.

Given space constraints, we present Chicago as the primary case in the manuscript and report results for the remaining metropolitan areas in the Online Appendix. Presenting all three research questions for one metropolitan area allows readers to observe the relationship between Geomarket composition (RQ1), group distributions (RQ2), and exclusion consequences (RQ3) within a single metropolitan structure. Chicago is well suited for this purpose. Its seven Geomarkets follow the canonical metropolitan structure described by Zensky and Oedel (1983), with a central city, first-ring suburbs, and outer suburbs. At the same time, Chicago exhibits substantial racial and socioeconomic differentiation across Geomarkets, making it useful for examining how Geomarket filters structure the composition of visible prospect pools. Chicago is also large enough to exhibit substantial variation while remaining sufficiently legible for readers to follow the full Geomarket structure.

Los Angeles and New York similarly exhibit strong differentiation across Geomarkets, but their large number of Geomarkets makes interpretation cumbersome in the main text. Other metropolitan areas, including Dallas-Fort Worth and the San Francisco Bay Area, follow alternative urban forms, such as bi-modal or metroplex structures with two centralized cores surrounded by suburban rings. Complete results for all metropolitan areas in our sample are provided in the Online Appendix. After presenting results for Chicago, we provide a high-level discussion of results from other metropolitan areas.

1.2 Data

The primary data source consists of student lists purchased through the College Board Student Search Service by public universities. We obtained these data through public records requests submitted to public universities in multiple states. The data sample for the broader project this

manuscript draws from includes a total of 414 orders associated with 2,549,085 prospects made by 11 public universities located in California, Illinois, Texas, and Arizona. Although these lists were purchased by individual universities, the set of prospects included in each list is a function of the search criteria specified for that student list purchase. A different university that selected the same set of search filters would obtain the same resulting prospect profiles. One caveat, most purchases select the option to exclude prospect profiles that were previously purchased by the university.

For each student list purchased from 2016 through 2020, we requested two related pieces of data: (1) the order summary, which specifies search criteria for the student list purchase; and (2) the de-identified prospect-level list produced from these search criteria. Prospect-level records contain information about students' high schools, geographic location, race/ethnicity, parental education, and other variables included in the purchased list. We use parental education to construct a first-generation indicator.

We also incorporated data from the 1980 Decennial Census, the 2000 Decennial Census, and the 2020 ACS 5-year estimates to provide descriptive historical and contemporary context about the racial and socioeconomic composition of Geomarkets.

Geomarket boundaries and prospect assignment

We digitized College Board Geomarket maps published in College Board (2023) and converted them into GIS shapefiles. In order to assign each each Census tract to a Geomarket, we georeferenced the College Board's Geomarket maps by overlaying PDF images onto spatial coordinates aligned with census tract boundaries. We then extracted vector data by manually tracing the Geomarket boundaries and generating individual shapefiles for each Geomarket. This approach to constructing Geomarkets aligned precisely with 2020 census tract boundaries from ACS 5-year estimates; that is, each census tract corresponded uniquely to a Geomarket generated from the College Board images. Finally, we spatially assigned prospect records to Geomarkets based on the geographic coordinates of their high school.

1.3 Analyses

We utilize descriptive statistics, tables, and maps to analyze how Geomarket filters structure the composition of visible prospect pools. First, we provide contextual descriptive analyses of Geomarkets themselves by describing the racial and socioeconomic differentiation across Geomarkets from 1980, 2000 and 2020.

Analyses for research questions 1-3 rely primarily on student list purchase data. For RQ1, we examine the racial/ethnic and first-generation composition of academically similar prospects – based on test score thresholds – within each Geomarket. These analyses show how Geomarket filtering partitions academically similar students into prospect pools that differ by race/ethnicity and parental education. For RQ2, we calculate the share of each racial/ethnic and first-generation group contributed by each Geomarket within a metropolitan prospect pool. These analyses identify Geomarkets that disproportionately contribute particular groups to the overall pool of academically eligible prospects. For RQ3, we show how the racial/ethnic and first-generation composition the prospect pool changes when specific Geomarkets are excluded from a student list purchase. These analyses estimate which groups become disproportionately invisible under alternative Geomarket selections.

2 Results

2.1 Geomarket socioeconomic and racial context

Figure X provides a static map of the seven “Greater Chicago” Geomarkets, as defined by College Board (2023). The Online Appendix provides an interactive version, showing change over time in population, race/ethnicity, poverty, and educational attainment at the tract- or Geomarket-level. The Geomarkets vary substantially in population. In 2020, the City of Chicago Geomarket had a population of about 3.2 million, followed by Western Suburbs at 1.4 million, South & Southwest Suburbs at 1.2 million. By contrast, the 2020 population of the North Shore and Evanston & Skokie Geomarkets was about 161,000 and 309,000, respectively.

Figure X provides Geomarket-level socioeconomic characteristics [REORDER FIGURES?]. Focusing on 2020, the North Shore Geomarket has the highest median income, followed by Evanston & Skokie,

Western Suburbs, Northwest Suburbs, and Chain of Lakes. At the opposite end of the spectrum, the City of Chicago has the lowest median income and the highest poverty rate, followed by the South & Southwest Suburbs, which has the lowest rate of BA+ educational attainment.

Figure X describes racial/ethnic composition. By 2020, Chicago-area Geomarkets varied sharply in racial/ethnic composition. The City of Chicago remained the most diverse Geomarket, with no racial/ethnic group constituting a majority. Several suburban Geomarkets that were predominantly White in 1980 had become more heterogeneous by 2020, especially the Chain of Lakes, Northwest Suburbs, Western Suburbs, and South and Southwest Suburbs. Much of this change reflected growth in Hispanic populations, alongside increases in Black residents in the South and Southwest Suburbs and Asian residents in the Northwest Suburbs and Evanston and Skokie. In 2020, Black and Hispanic residents together accounted for 55.4% of the City of Chicago Geomarket and 43.0% of the South and Southwest Suburbs Geomarket. North Shore remained comparatively White.

2.2 Research Question 1

RQ1 asks, to what extent do College Board Geomarkets partition academically similar prospects into racially and socioeconomically distinct pools within metropolitan areas? Figure X shows the racial/ethnic composition of purchased student profiles by Geomarket across three different SAT score ranges, 1020 - 1150, 1160 - 1300, and 1310 - 1600. Focusing on the middle test score range, the racial/ethnic composition of prospect pools varied sharply across Chicago-area Geomarkets. White prospects were most concentrated in North Shore, Evanston & Skokie, and Western Suburbs, while Asian prospects were especially concentrated in Evanston & Skokie, Northwest Suburbs, and North Shore. Black and Hispanic prospects were concentrated most heavily in the City of Chicago and South & Southwest Suburbs. Asian representation increases especially sharply in North Shore and Western Suburbs at the highest score range. However, the broad takeaway is that racial/ethnic differences in Geomarket composition persist across score ranges.

Figure X shows the first-generation status composition of Geomarkets across the three score ranges. Predictably, the share of first-generation students declines as test score thresholds rise. Nevertheless, we observe consistent and substantial differences in the share of first-generation students across Geomarkets. Focusing on the middle score range, first-generation prospects account for approximately

47% of profiles in the City of Chicago and 38% in South & Southwest Suburbs, compared to 22% in Western Suburbs, 19% in Evanston & Skokie, and 11% in North Shore. Northwest Suburbs and Chain of Lakes occupy intermediate positions, at 29% and 24%, respectively. This pattern is consistent with the socioeconomic characteristics described above, as Geomarkets with lower median income, higher poverty, and lower BA+ attainment contain substantially larger shares of first-generation prospects.

Figure X presents first-generation status by race for the middle test score range for White, Asian, Black, and Hispanic students. Figure X presents first-generation status by race for the middle test score range. The main takeaway is that first-generation status varies substantially across Geomarkets even within the same racial/ethnic group. These within-race differences are largest for Asian and Hispanic prospects and smaller for White and Black prospects. Among Asian prospects, more than 50% of students from the City of Chicago are first-generation, compared to roughly 20% in Western Suburbs. Among Hispanic prospects, approximately 75% of students from the City of Chicago are first-generation, compared to about 25% in North Shore. Thus, Geomarkets do not merely separate racial/ethnic groups from one another; they also distinguish socioeconomically different subsets within racial/ethnic groups.

2.3 Research Question 2

RQ2 asks which Geomarkets contribute disproportionately to different racial/ethnic and first-generation groups within academically similar prospect pools. Figure X shifts attention from the composition of each Geomarket to where different groups come from within the Chicago metropolitan prospect pool. The lollipop lines show each Geomarket's share of a given racial/ethnic group, while the grey bars show its share of all prospects; lines above the grey bars indicate overrepresentation, and lines below indicate underrepresentation.

In Figure X, White prospects come mostly from suburban Geomarkets, especially Western Suburbs, South & Southwest Suburbs, Chain of Lakes, and Northwest Suburbs, while City of Chicago contributes disproportionately fewer White prospects. Asian prospects are concentrated in Western Suburbs, City of Chicago, Northwest Suburbs, and Evanston & Skokie, with North Shore also contributing disproportionately more Asian prospects. Black prospects are disproportionately

concentrated in City of Chicago and South & Southwest Suburbs, with the more affluent suburban Geomarkets contributing disproportionately fewer Black prospects. Hispanic prospects also come disproportionately from City of Chicago, with substantial contributions from Western Suburbs, South & Southwest Suburbs, Chain of Lakes, and Northwest Suburbs. Overall, Figure X shows that selecting or excluding a Geomarket does not simply change the size of the prospect pool; it also changes which racial/ethnic groups are disproportionately made visible or invisible within that pool.

Figure X shows the contribution of each Geomarket to the overall share of first-generation prospects and the overall share of non first-generation prospects. The City of Chicago stands out as an engine for first-generation prospects, contributing 23% of all prospects but 35% of first-generation prospects. Similarly, South & Southwest Suburbs contributes 17% of all prospects but 21% of first-generation prospects. By contrast, all other Geomarkets contribute disproportionately fewer first-generation prospects and disproportionately more non-first-generation prospects relative to their overall share of the prospect pool. The North Shore and Western Suburbs Geomarkets stand out as the clearest examples, contributing relatively large shares of non-first-generation prospects and relatively small shares of first-generation prospects.

Figure X examines Geomarket contributions by race and first-generation status. The gray bar shows the share of all prospects who come from that Geomarket. The blue lollipop shows the share of first-generation prospects from a given racial/ethnic group who come from that Geomarket, while the red lollipop shows the corresponding share for non-first-generation prospects from the same racial/ethnic group. Whereas the gray bars show which Geomarkets contribute substantial shares of the overall prospect pool, the lollipops show where race-by-first-generation subgroups depart from the overall pattern. Among White prospects, first-generation students are drawn more heavily from the South & Southwest Suburbs, while non-first-generation students are drawn more heavily from the Western Suburbs. Chicago, by contrast, contributes a smaller share of both first-generation and non-first-generation White prospects than it does of the overall prospect pool. Among Black prospects, both first-generation and non-first-generation students are concentrated overwhelmingly in the City of Chicago and South & Southwest Suburbs, with relatively small contributions from the remaining Geomarkets. Thus, for Black prospects, Geomarkets strongly structure where students come from, but they do not substantially distinguish first-generation from non-first-generation

students.

Asian prospects exhibit a different pattern. First-generation Asian prospects are drawn especially heavily from the City of Chicago, which contributes a much larger share of first-generation Asian prospects than of the overall prospect pool. Non-first-generation Asian prospects, by contrast, are drawn more heavily from the Western Suburbs and Northwest Suburbs. This pattern is not uniform, however; Evanston & Skokie contributes disproportionately larger shares of both first-generation and non-first-generation Asian prospects than it does of the overall prospect pool. Hispanic prospects show a similarly sharp first-generation split. First-generation Hispanic prospects are concentrated especially heavily in the City of Chicago, while non-first-generation Hispanic prospects are more broadly distributed across suburban Geomarkets. Overall, Figure X suggests that Geomarkets operate differently across racial/ethnic groups. For White and Black prospects, some Geomarkets contribute much larger shares of students than others, but first-generation and non-first-generation students within each group are distributed relatively similarly across those Geomarkets. For Asian and Hispanic prospects, by contrast, first-generation status is more geographically patterned, with first-generation students drawn especially heavily from the City of Chicago and non-first-generation students drawn more heavily from suburban Geomarkets.

2.4 Research Question 3

RQ3 asks how excluding particular Geomarkets changes the composition of the prospect pool made visible to colleges. Figures X and Y answer this question for race/ethnicity. Each point represents a counterfactual exclusion scenario, showing how the racial/ethnic composition of the prospect pool would change if that Geomarket were excluded from the student-list purchase. Figure X shows changes in percentage points. For example, White students represent 60.4% of the prospect pool when all Greater Chicago Geomarkets are included in the purchase. Excluding the Western Suburbs Geomarket would produce a 3.1 percentage-point reduction, such that White students would comprise $60.4\% - 3.1\% = 57.3\%$ of the remaining prospect pool. By contrast, excluding the City of Chicago Geomarket would produce a 7.6 percentage-point increase, such that White students would comprise $60.4\% + 7.6\% = 68.0\%$ of the remaining prospect pool.

Black students represent 7.0% of the prospect pool across all Geomarkets. Excluding the City of

Chicago Geomarket would produce a 2.6 percentage-point reduction, such that Black students would comprise $7.0\% - 2.6\% = 4.4\%$ of the remaining prospect pool. However, this 2.6 percentage-point reduction from Figure X is associated with a much larger relative percent change. Figure Y shows the same counterfactual exclusions in relative terms, calculated as the remaining-pool share minus the full-pool baseline share, divided by the full-pool baseline share. Here, excluding the City of Chicago would reduce Black students' share of the remaining prospect pool by approximately 36% relative to their full-pool baseline: $((4.4\% - 7.0\%)/7.0\%) \times 100 = -37.1\%$. [CRYSTAL – CHECK NUMBERS]. Similarly, excluding the South & Southwest Suburbs would reduce Black students' share of the remaining prospect pool by 0.8 percentage points, from 7.0% to 6.2%, an 11% decline relative to their full-pool baseline. By contrast, excluding the Western Suburbs would increase Black students' share of the remaining prospect pool by 1.4 percentage points, from 7.0% to 8.4%, a 20% increase relative to their full-pool baseline.

Hispanic students represent 18.8% of the prospect pool across all Geomarkets. Excluding the City of Chicago Geomarket would reduce Hispanic students' share of the remaining prospect pool by 5.2 percentage points, from 18.8% to 13.6%, a 28% decline relative to their full-pool baseline. By contrast, excluding the Western Suburbs would increase Hispanic students' share of the remaining prospect pool by 2.0 percentage points, from 18.8% to 20.8%, a 10% increase relative to their full-pool baseline. Taken together, the race/ethnicity results show that excluding the City of Chicago would not merely remove a large Geomarket from the purchase. It substantially changes who remains visible to colleges, increasing the share of White prospects while reducing the shares of Black and Hispanic prospects.

Figures X and Y repeat the counterfactual exclusion analysis by first-generation status. Figure X reports percentage-point changes. First-generation students comprise 30.5% of the prospect pool across all Geomarkets, but a disproportionate share of these students come from the City of Chicago and South & Southwest Suburbs Geomarkets. Excluding the City of Chicago would reduce the first-generation share by 4.8 percentage points, from 30.5% to 25.7% of the remaining pool. Figure Y reports these results in terms of relative percent change; excluding the City of Chicago would produce a 16% decline in the first-generation share relative to the full-pool baseline: $((25.7 - 30.5)/30.5) \times 100 = -15.7\%$. Similarly, excluding South & Southwest Suburbs would

reduce the first-generation share by 1.5 percentage points, from 30.5% to 29.0%, representing 5% decline relative to the full-pool baseline. By contrast, excluding Western Suburbs would increase the first-generation share by 3.5 percentage points, from 30.5% to 34.0%, a 12% increase relative to the full-pool baseline. Results for non-first-generation students move in the opposite direction. Excluding Western Suburbs or North Shore would reduce the non-first-generation share of the remaining pool, while excluding the City of Chicago or South & Southwest Suburbs would increase it.

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